

10. a) In the math formula, the second part depends on correlation (ρ). If ρ is low or negative, the " $w_i * w_j * \sigma_i * \sigma_j * \rho_{ij}$ " part becomes smaller or even negative. This shrinks the total sum. So, the total portfolio variance (risk) goes down. This means the assets do not all crash at the same time.

b) I would add the P/E (Price to Earnings) ratio. This tells us if a stock is too expensive based on real company profits, not just past prices. To get it, I would download quarterly earnings data from a finance API. Then, I would fill in the daily P/E ratio values and add it as a new column in Pandas before normalizing the data.

c) If you add too many features, the math space gets too huge. In high dimensions, the distance between any two points becomes almost exactly the same. So, finding the nearest neighbour stops making sense. This ruins the model's accuracy. To fix this, you can use PCA to shrink the number of features before running KNN.

d) The three causes are:

1. Look-ahead bias (using future data you would not have at the time).
2. Survivorship bias (only testing on companies that survived and ignoring bankrupt ones).
3. Data snooping (testing hundreds of rules until one looks good by pure luck). To detect this gap, you must use out-of-sample testing. This means hiding a chunk of new data during training and only testing the final model on it.

e) 1. Volatility clustering: Financial data has random periods of high and low risk. This breaks the rule of constant variance (heteroskedasticity).

2. Autocorrelation: Today's return is often linked to yesterday's return, so they are not totally independent. These violations make the model overconfident. The model will output standard errors that are too small, making you trust a bad prediction.